

MEDASSIST-AI

Milestone-2 Report

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Model Training

1. Project Description

MedAssist-AI: Medical Symptom Analysis and Disease Prediction System

MedAssist-AI is an Artificial Intelligence (AI) powered healthcare platform designed to assist users in analyzing their symptoms, predicting possible diseases, assessing health risks, and providing preliminary healthcare recommendations. The primary objective of the system is to help users understand potential health conditions at an early stage and support informed healthcare decisions through AI-driven medical insights.

The platform is designed to support various healthcare functionalities, including symptom analysis, disease prediction, risk assessment, treatment recommendations, health report generation, and patient history management. It can be utilized by patients, healthcare providers, clinics, hospitals, and telemedicine platforms as an intelligent decision-support system.

The project integrates multiple healthcare datasets, such as the Disease Symptoms and Patient Profile Dataset, Disease Prediction Using Symptoms Dataset, and the CDC Behavioral Risk Factor Surveillance System (BRFSS) dataset, to build a comprehensive disease prediction model. These datasets provide information related to symptoms, diseases, patient profiles, lifestyle indicators, and healthcare conditions, enabling the system to make reliable predictions.

The complete MedAssist-AI platform consists of several modules, including User Management, Symptom Analysis, Disease Prediction, Risk Assessment, Treatment Recommendation, Health Reports, and Analytics Dashboard. Together, these modules provide an end-to-end intelligent healthcare solution.

2. Objectives

- Train Decision Tree classifier.
- Train Random Forest classifier.
- Evaluate both models.
- Compare their performance.

3. Decision Tree Model Development

The Decision Tree algorithm was implemented as the first classification model in the MedAssistAI project. It is a supervised machine learning algorithm that predicts the target class by learning a series of decision rules from the training data. The model recursively divides the dataset into smaller subsets based on the most informative symptom features, creating a tree-like structure that is easy to interpret.

The model was trained using the preprocessed dataset consisting of 377 symptom features and 773 disease classes. The dataset was divided into training and testing sets using an 80:20 split. After successful training, the Decision Tree model was saved using the Joblib library for future predictions and evaluation.



```
print("Training Decision Tree...")

dt_model.fit(x_train, y_train)

print("Decision Tree Trained Successfully!")
```

[7]

```
... Training Decision Tree...
Decision Tree Trained Successfully!
```

Figure 3.1: *Decision Tree Model Training*

4. Random Forest Model Development

The Random Forest algorithm was implemented as the second classification model in the MedAssistAI project. It is an ensemble learning algorithm that consists of multiple Decision Trees, where each tree is trained on a random subset of the training data. The final prediction is obtained through majority voting among all the individual trees, which generally improves the model's accuracy and reduces the risk of overfitting compared to a single Decision Tree.

Initially, the Random Forest model was trained on the complete preprocessed dataset containing 189,647 records and 377 symptom features. However, due to the large dataset size and the available system memory of 8 GB RAM, the training process resulted in a MemoryError, making it impossible to train the model on the full dataset. To overcome this limitation, a representative sample of the dataset was used for training the Random Forest model. Although the sampled dataset affected the model's overall performance, it successfully demonstrated the implementation and evaluation of the Random Forest algorithm within the available hardware constraints. The trained model was then saved using the Joblib library for further evaluation and comparison.

```
> \n\n    print("Training Started...")\n\n    rf_model.fit(X_train, y_train)\n\n    print("Model Trained Successfully!")\n\n[8]\n\n.. Training Started...\n   Model Trained Successfully!
```

Figure 4.1: *Random Forest Model Training*

5. Model Evaluation

After successfully training the Decision Tree and Random Forest models, their performance was evaluated using standard classification metrics. Since the MedAssistAI dataset contains 773 disease classes, relying solely on accuracy was not sufficient. Therefore, additional evaluation metrics such as Precision, Recall, and F1-Score were used to obtain a comprehensive assessment of the models' predictive performance. These metrics provide a better understanding of how effectively each model classifies diseases while considering the class imbalance present in the dataset.

In addition to these evaluation metrics, classification reports were generated for both models to summarize their performance across all disease classes. Furthermore, feature importance analysis was performed to identify the symptoms that contributed most significantly to the prediction process. Finally, a comparative analysis was carried out by visualizing the performance of both models using tables and graphs, making it easier to analyze their strengths and limitations.

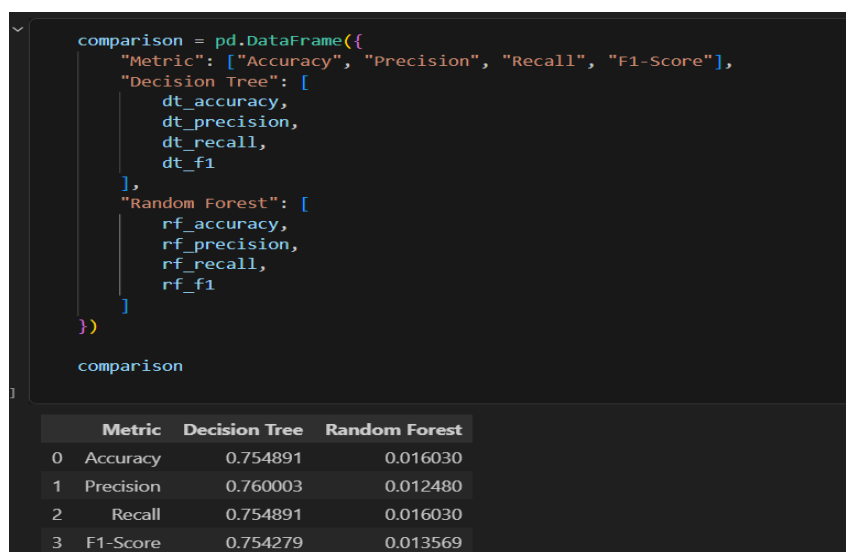


Figure 5.1: *Performance Metrics of Decision Tree and Random Forest Model*

5.1 Decision Tree Evaluation

The performance of the Decision Tree model was evaluated using the testing dataset after the training process was completed. Standard classification metrics, including Accuracy, Precision, Recall, and F1-Score, were calculated to measure the effectiveness of the model in predicting diseases based on patient symptoms. In addition to these metrics, a detailed classification report was generated to analyze the model's performance across all disease classes.

The evaluation results indicated that the Decision Tree model achieved satisfactory performance and was able to classify diseases effectively using the available symptom data. The generated classification report provides a detailed summary of the model's predictive performance and serves as an important basis for comparison with other machine learning models.

```
print("=" * 60)
print("DECISION TREE CLASSIFICATION REPORT")
print("=" * 60)

print(classification_report(
    y_test,
    dt_pred,
    zero_division=0
))
```

	precision	recall	f1-score	support
0	0.27	0.60	0.38	5
1	0.87	0.92	0.89	49
2	0.39	0.66	0.49	29
3	1.00	1.00	1.00	1
4	0.41	0.76	0.54	25
5	0.25	0.50	0.33	2
6	0.00	0.00	0.00	1
7	0.00	0.00	0.00	4
8	0.46	0.63	0.53	82
9	0.49	0.68	0.57	155
10	0.83	0.91	0.87	238
11	0.62	0.67	0.65	247
12	0.51	0.64	0.57	166
13	0.00	0.00	0.00	0
14	0.11	0.23	0.15	30
15	0.86	0.90	0.88	168
16	0.57	0.60	0.59	108
17	0.76	0.89	0.82	243
18	0.70	0.77	0.73	57
19	0.69	0.82	0.75	170
...				
accuracy			0.75	37930
macro avg	0.61	0.61	0.59	37930
weighted avg	0.76	0.75	0.75	37930

Figure 5.2: *Decision Tree Classification Report*

5.2 Random Forest Evaluation

The Random Forest model was evaluated using the same performance metrics as the Decision Tree model to ensure consistency in comparison. Accuracy, Precision, Recall, F1-Score, and the classification report were generated using the testing dataset. These evaluation metrics helped in assessing the predictive capability of the Random Forest classifier.

Although the Random Forest algorithm generally provides better generalization by combining multiple decision trees, its performance in this project was comparatively lower

because the model was trained on a sampled dataset. This approach was adopted due to hardware limitations that prevented training on the complete dataset. Consequently, the evaluation results should be interpreted considering the reduced training data rather than the algorithm's actual potential.

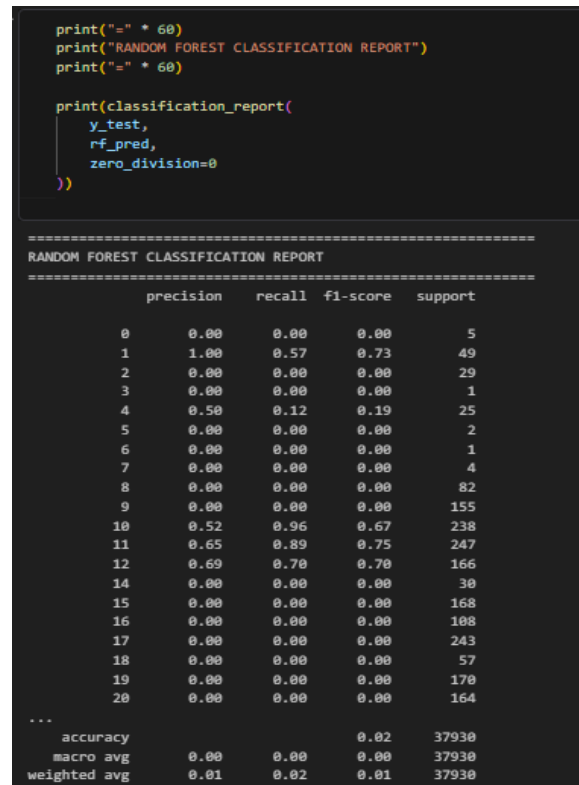


Figure 5.3: *Random Forest Classification Report*

6. Conclusion

In this milestone, the Decision Tree and Random Forest machine learning models were successfully implemented for disease prediction using symptom-based data. Both models were trained, evaluated, and analyzed using standard classification metrics, including Accuracy, Precision, Recall, and F1-Score. Feature importance analysis was also performed to identify the symptoms that contributed most significantly to the prediction process.

The implementation of these models enhanced the understanding of classification algorithms and their practical application in disease prediction. Despite the hardware limitations encountered during Random Forest training, both models were successfully implemented and evaluated, contributing to the overall development of the MedAssistAI project.